Do 11,000 Sharks Die Every Hour?

MA26M017

How unreported catches and finning drive staggering global mortality.



METHODS

Converted reported and estimated catch tonnages using specific and overall median weights.

KEY RESULT

Estimated 100 million global shark deaths in the year 2000.

TAKEAWAY

Fishing exploitation severely exceeds natural population rebound limits.

Media: BBC More or Less (Sep 2025) | **Data:** Worm et al. (2013)

Timestamps: 01:10 (Key Result) • 01:40 (Variety of Sharks) • 02:26 (Extinction Risk) • 02:55 (Methods) • 03:50 (Estimation) • 05:34 (Limitations)

Why are so many ex-prisoners being sent back to prison?

In England and Wales, prison recalls increased to 51,419 in the year ending March 2026, a 28% rise compared with the previous year. Many prisoners are recalled not for committing new crimes but for breaking release conditions. We aim to investigate the reasons behind this increase using hypothesis testing and logistic regression.

Objective: To investigate whether reoffending is associated with age group

- To test this hypothesis, the chi-square test of independence is carried out
- P-value=0.37
- Conclusion: Since P-value is greater than usual level of significance

There is insufficient statistical evidence of an association between age group and reoffending

Objective: To investigate whether reoffending is associated with housing condition.

- To test this hypothesis, the chi-square test of independence is carried out
- P-Value= 8.4×10^{-17}
- Conclusion: Since the P-value is smaller than usual level of significance

There is sufficient statistical evidence of an association between housing condition and reoffending

Conclusion: Housing condition, sentence length, and previous offences were significantly associated with reoffending. Homeless prisoners had approximately 5 times higher odds of reoffending, while each additional previous offence increased the odds by about 2.2%. Sentence lengths of 12 months or more were associated with lower odds of reoffending. However, age and gender were not statistically significant predictors in the model.

Objective: To study how different factors affect the probability of a prisoner reoffending after release.
For this purpose, we used a logistic regression model to examine how different factors are related to the probability of reoffending after release
The logistic regression model is as follows,

$$\ln\left(\frac{\hat{p}_i}{1-\hat{P}_i}\right) = 0.3951 - 0.0197G_i - 0.0028A_{1i} - 0.1862A_{2i} - 0.0922A_{3i} - 0.1456A_{4i} - 1.6313S_i + 1.6122H_i$$

Explanatory variable	Coefficient	Odds ratio	Interpretation
Female vs Male	-0.0197	0.981	1.9% lower odds
Age 21–30 vs 18–20	-0.0028	0.997	0.3% lower odds
Age 31–40 vs 18–20	-0.1862	0.83	17.0% lower odds
Age 41–50 vs 18–20	-0.0922	0.912	8.8% lower odds
Age 50+ vs 18–20	-0.1456	0.865	13.5% lower odds
Sentence 12+ months vs less than 12 months	-1.6313	0.196	80.4% lower odds
Previous offences (per additional offence)	0.02195	1.022	2.2% higher odds
Homeless vs Stable housing	1.6122	5.014	401.4% higher odds

CAN 8,000 TRACKED WEALTHY PEOPLE PREDICT 9,500 DEPARTING UK MILLIONAIRES?

Presented by: B Charulekha | NA23B041

News headline Rich ready to leave UK over budget tax threat

602,000 UK Millionaires → 9500 estimated departures



Observed dataset (Sample)

~ 50 net departures in Jan-May (5 months)

50 in 8,000 = 0.625%

Extrapolating to UK Millionaires population size (Jan-May)

UK Millionairs population = 602,000

0.625% of 602,000 = 3762.5

Extrapolating for the whole year of 2024

5 Months = 3762.5

for 12 Months, 3762.5 * 12 / 5 = 9,030

METHODS Track LinkedIn location changes in 8,000 people sample, extrapolate to 602,000 UK Millionares

KEY RESULT About 50 net leavers is projected to 9,500 annual departures from UK

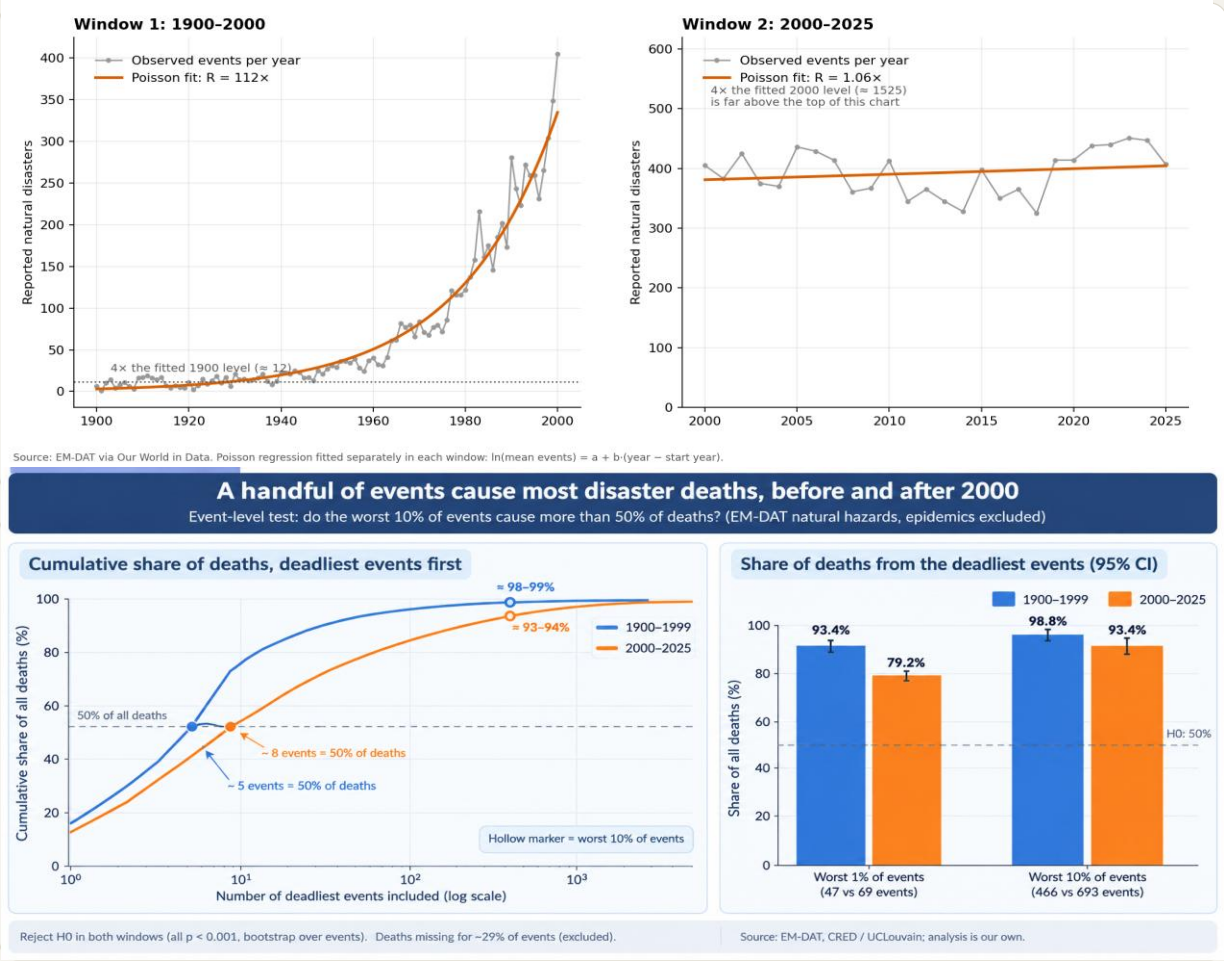
DATASET ISSUES Sample is skewed towards the ultra-rich population | Labour law changes causation is unsupported

How do you count millionaires? | More or Less | BBC | 18 Sept 2024

Relevant timestamps : Database [02:44] | UK millionaire population [02:17] | Net departure [03:18] | Data collection method [02:50] | Dataset issues [03:38]

Did Natural Disasters Really Quadruple?

We tested the claim on EM-DAT data.



Hypothesis 1 : Is the Rise in Recorded Disasters Truly Fourfold or more?

H_0: R ≤ 4 H1: R>4 ; Test: Wald z test on the slope of quasi poisson regression; α=0.05

1900–2000

Reject H_0

R = 112x

95% CI : (86 , 146)x

z = 24.6 , p < 0.001

2000–2025

Fail to reject H_0

R = 1.06×

95% CI: (0.94, 1.20)x

z = −20.9 , p ≈ 1

NOTE: EM-DAT(established in 1988) warns pre-2000 data is subject to reporting biases.

Hypothesis 2 : Do the deadliest 10% of disasters account for more than 50% of total deaths?

H_0: S<=0.5 H1: S>0.5 ; Test: Bootstrap hypothesis test **; α=0.05**

1900–1999

Reject H_0

98.8% - recorded deaths

95% CI: (97.6-99.3)%

2000–2025

Reject H_0

93.4% - recorded deaths

95% CI:(89.5,95.5)%

1900-1999: China's flood in 1931 caused 3.7M deaths

2000-2025: Haiti Earthquake in 2010 caused 223k deaths

Key result: Since 2000, we find no evidence of a fourfold increase in recorded disasters
-- A few extreme events drive disaster deaths.

Takeaway- The fourfold claim depends on the start year.

Source: BBC Radio 4, More or Less: Behind the Stats, [Title: "Are natural disasters on rise?"]. Data: EM-DAT (CRED / UCLouvain) via Our World in Data; tests are our own.

DO DIABETES-RELATED HEALTH INDICATORS SHOW THE SAME PATTERNS ACROSS TWO CATEGORIES OF INDIVIDUALS?

Comparing Age, Glucose and BMI using logistic regression and interaction testing

METHODS & MODELS

Two datasets: Pima Indians Diabetes and Indians Diabetes
Outcome (Y): diabetes status (0 = no, 1 = yes)
D: dataset indicator (0 = Pima, 1 = Prediction)

Reduced model (no interactions)

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 D + \beta_2 A + \beta_3 G + \beta_4 B$$

Full model (with interactions)

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 D + \beta_2 A + \beta_3 G + \beta_4 B + \beta_5 (D \times A) + \beta_6 (D \times G) + \beta_7 (D \times B)$$

A = standardized Age
G = standardized Glucose
B = standardized BMI
p = probability of diabetes

HYPOTHESES

Null hypothesis (H_0)

The Age, Glucose and BMI associations with diabetes are the same across both categories.

$$\beta_5 = \beta_6 = \beta_7 = 0$$

Alternative hypothesis (H_1)

At least one of these associations differs between the two categories.

$$\text{At least one of } \beta_5, \beta_6, \beta_7 \neq 0$$

Test: Likelihood Ratio Test comparing the reduced and full models (χ^2, 3 df).

RESULTS

Overall likelihood ratio test

p = 0.0236

Reject H_0 at α = 0.05

Individual interaction tests

Dataset × Age	**0.0022**	**Significant**
Dataset × Glucose	**0.4967**	**Not significant**
Dataset × BMI	**0.9317**	**Not significant**

Takeaway: Age shows the clearest individual difference. Glucose and BMI interactions are not individually significant.

Interpretation: evidence of differing associations between these datasets; this does not establish causation.

Inspired by: “Tale of Two Indians: Heterogeneity in Type 2 Diabetes Pathophysiology” (2019).

Are Refugees more likely to commit Crime ?

"Asylum seekers make up 0.8% of Dorset's population and 44% of alleged sex offenses "

Data Source	Offender	Claimed Rate	Actual
Social Media Post	Asylum Seekers	**44%**	Fabricated
Dorset Police FOI	Migrant Hotel Residents	N/A	**~1.0%(8 of 808 reported)[1]**
National Level FOI	Foreign Nationals	N/A	**7.0% - 8.0%**

Grok AI

" Dorset Police's FOI disclosure confirms that 51 of 116 males awaiting prosecution for sexual offenses in the Bournemouth area are asylum seekers from migrant hotels, which calculates to about 44% " [2]

KEY RESULT

1) **Viral 44% claim fabricated**; verified police data indicates 1%.

2) **AI Hallucination :** AI cited fake document to claim 44%. Admitted non-existence of data upon further prompting.

3) Most refugees from Afghanistan, Syria, Iran, and Eritrea come to the UK. They usually arrive through government programs/asylum system. **War is the main reason.**

4) Demographic skew as most asylum seekers are young

Timestamps
(1) - 1:41 (Investigated for crimes), (2) - 3:48 (Conversation with Grok), (3) -5:16 , 6:04 (Conversation with Grok)

Alcohol and heart health: hypotheses and results

A: Blood Pressure

H_0 : $\mu_{\text{drinkers}} = \mu_{\text{non-drinkers}}$

H_1 : $\mu_{\text{drinkers}} > \mu_{\text{non-drinkers}}$

Systolic BP. One-sided Welch t-test (unequal variance)

B: Heart Disease

H_0 : $p_{\text{drinkers}} = p_{\text{non-drinkers}}$

H_1 : $p_{\text{drinkers}} \neq p_{\text{non-drinkers}}$

Heart disease. Two-proportion z-test ($n = 68{,}610$)

C: Only Non-smokers

H_0 : $p_{\text{drinkers}} = p_{\text{non-drinkers}}$

H_1 : $p_{\text{drinkers}} \neq p_{\text{non-drinkers}}$

Same z test as B on non-smokers ($n = 62{,}574$),

	Drinkers	Non-drinkers	Result ($\alpha = 0.05$)
A: mean systolic BP(mmHg)	129.2	126.5	$\lvert T_s\rvert = \lvert 8.68\rvert > 1.96,\ p < 0.001 < 0.05$
B: share with heart disease	47.7%	49.6%	$\lvert Z_s\rvert = \lvert -2.245\rvert > 1.96,\ p = 0.024 < 0.025$
C: heart disease in non-smokers	51.3%	49.7%	$\lvert Z_s\rvert = \lvert 1.421\rvert < 1.96,\ p = 0.155 < 0.025$

Conclusion.

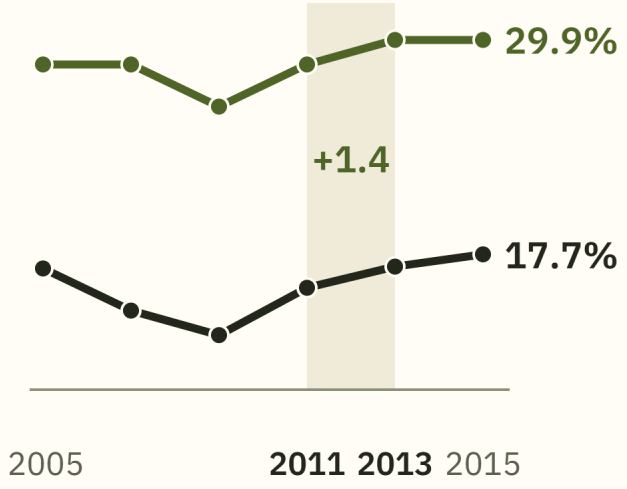
- Drinkers have higher systolic BP (reject H_0). It holds among non-smokers too.
- Drinkers show slightly less heart disease (reject H_0). This does not mean alcohol protects the heart.
- Among non-smokers, drinkers have slightly more heart disease(fail to reject H_0).
- Alcohol drinking is significantly associated with higher blood pressure, but no reliable evidence for changes the heart disease rate. Drinking is a yes/no with no amounts so there are association, not proof of cause.

Cardiovascular Disease dataset (OpenML 45547), about 68,600 records after cleaning.

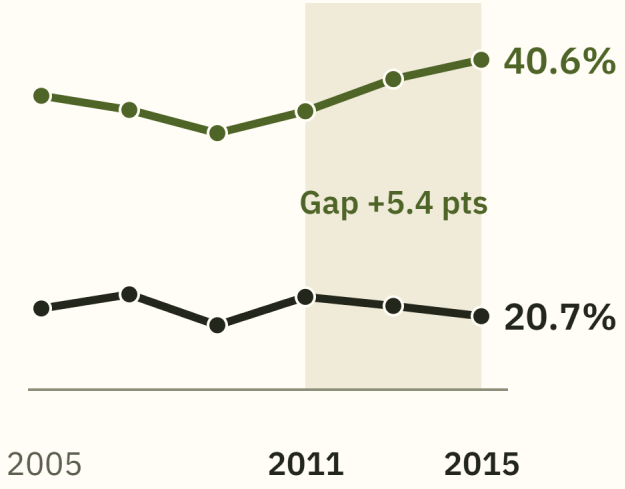
Did social media cause the teen mental health crisis?

Teen hopelessness rose gradually, mostly among girls; is social media the cause?

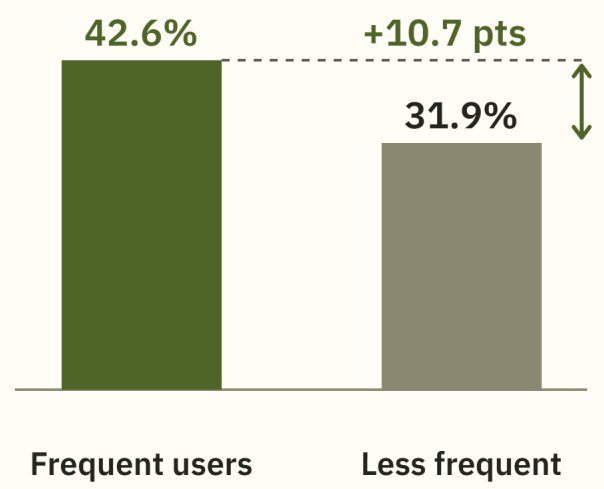
1 · Cliff or slow climb?



2 · Girls vs boys, felt hopeless



3 · Heavy vs lighter use



METHODS

Z-tests and difference-in-differences on CDC survey data; prevalence ratio for 2023.

KEY RESULT

Gradual rise; girls +3.95 vs boys −1.45; heavy users 42.6% vs 31.9%.

TAKEAWAY

Gradual, girl-driven rise; social media linked, but cause unproven.

"Jonathan Haidt Explains the Youth Mental Health Crisis," The Screenagers Podcast, Jun 10, 2024 · 03:52 · 05:57 · 09:04

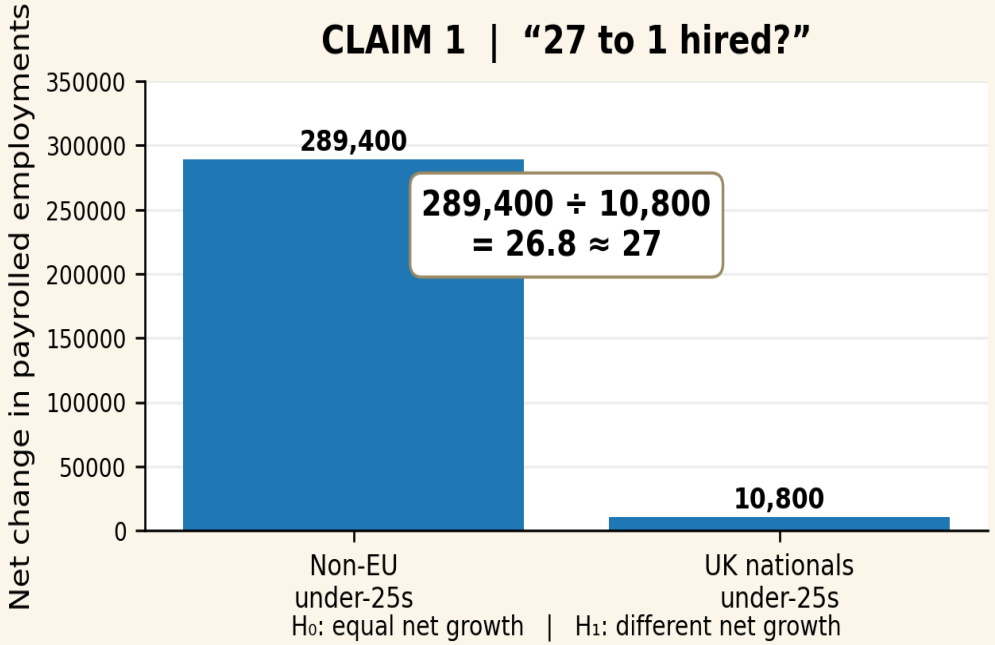
Data: CDC Youth Risk Behavior Survey (cdc.gov/yrbs)

WHEN HEADLINES MEET DATA: TWO STATISTICAL CHECKS

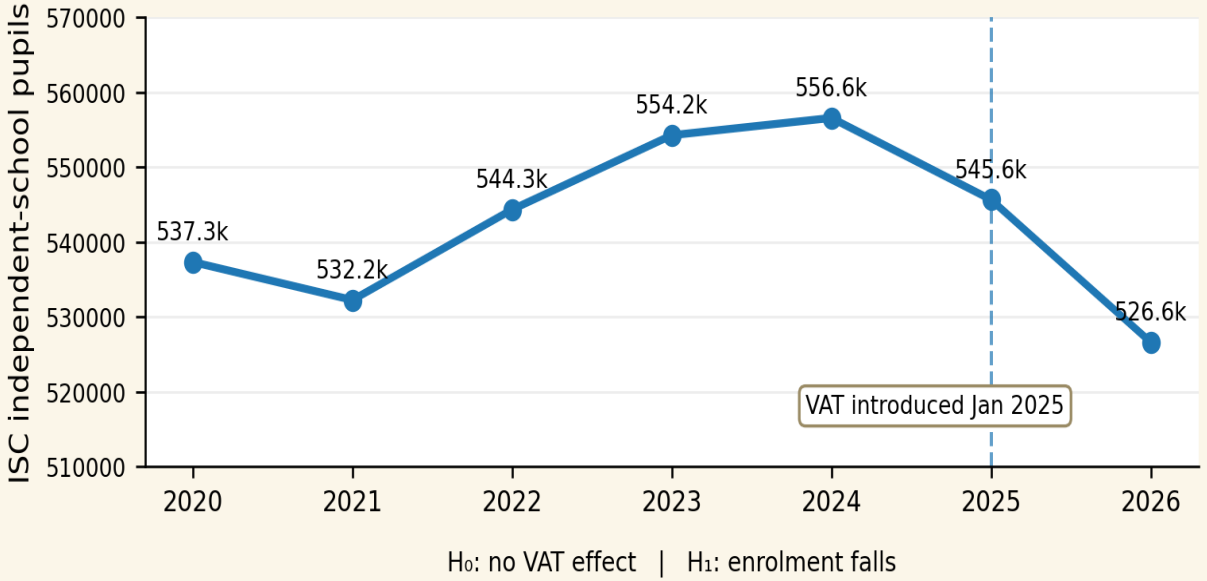
A number can be correct—and still tell the wrong story.

BBC Radio 4 | More or Less: "Education, Education, Education (and immigration)" | 10 June 2026

CLAIM 1 | "27 to 1 hired?"



CLAIM 2 | "Did VAT reduce enrolment?"



METHODS

Compare payroll growth; test VAT impact using Difference-in-Differences.

KEY RESULT

Ratio equals 26.8; pupil totals fell 1.96% after 2024.

TAKEAWAY

Net job growth is not hires; trends alone prove no causation.

LIMIT: descriptive figures only; matched population data and state-school comparisons are needed for formal tests.

SOURCE: BBC Radio 4, More or Less, "Education, Education, Education (and immigration)," 10 Jun 2026; 00:32 (employment), 08:40 (VAT). Data: HMRC payroll figures (CSJ analysis); ISC Census 2020–2026. 95% CI/DiD are proposed, not completed tests.

STATISTICS & INFERENCE | BBC RADIO 4: MORE OR LESS

THE STATS OF THE NATION

How do we evaluate claims about immigration, benefits and inequality?

DESCRIPTIVE STATISTICS → summarise observed data | STATISTICAL INFERENCE → assess what the data supports

1. IMMIGRATION

98% claim

Descriptive: calculate the share of population growth associated with net migration.

Inference: check the population estimates, time period and uncertainty before generalising.

2. BENEFIT SPENDING

0.1% of GDP

Descriptive: express unemployment-benefit spending as a proportion of GDP and compare countries.

Inference: ask whether countries use comparable definitions and datasets.

3. INCOME INEQUALITY

14.5% top 1% share

Descriptive: measure the income share of the richest 1%; compare it over time.

Inference: consider data coverage, capital gains and whether the trend is robust.

A SIMPLE INFERENCE CHECK

Claim → define the measure → identify the data/sample → quantify uncertainty → draw a cautious conclusion

Source: BBC Radio 4, More or Less, “The Stats of the Nation” (9 Jan 2026). Figures are episode-reported and depend on definitions/time periods.

MA 5104 | STATISTICAL INFERENCE & LEARNING

ARE INDIANS REALLY HAPPY?

India’s rank is low—but what do the numbers really reveal?

1 | WHAT WAS MEASURED?

4.389 / 10

India’s mean life evaluation

RANK 118 of 147 countries

0 = worst life | 10 = best life

2022–2024 average • Gallup World Poll

95% rank interval: 114–129

2 | INDIA: INDICATORS REPORTED IN ARTICLE

23rd	Personal freedom
93rd	GDP per capita
56th	Perceived corruption
60.6 y	Healthy life expectancy
58%	Have someone to rely on

Article-reported context; not all determine the rank.

3 | HOW THE ANALYSIS WORKS

GALLUP SURVEY → CANTRIL LADDER

Gallup asks people to rate their life from 0–10.

3-YEAR AVERAGE (2022–24)

Uses a pooled 3-year average (2022–24).

CROSS-COUNTRY OLS REGRESSION

Models six factors: income, support, health, freedom, generosity and corruption.

4 | REGRESSION EVIDENCE (WHR 2025, TABLE 2.1)

Outcome: Cantril Ladder score | Pooled OLS, 155 countries, 2,234 observations

Social support	+2.686	**Freedom**	+1.518	**GDP**	+0.328

Associations, not proof of cause and effect.

5 | INTERPRETATION & LIMITATION

A national average hides differences across people and regions.

A survey score cannot capture every lived experience.

India's happiness ranking is not meaningless, but neither is it the complete story of Indian happiness.

Sources: World Happiness Report 2025, Fig. 2.1 & Table 2.1 (Gallup World Poll); Maya Pillai, “India World Happiness Report 2025: Are Indians Really Unhappy?”